

**On Geometric Origins, Spatial
Transformations Between of Spaces of Infinite
Dimensional Vectors In Which the Vector's
Dimensions are complex, and N -Dimensional
Spaces.**

Draft 2

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Introduction

Here I will be explaining a geometric system for transformations between geometric spaces with varying dimensions. I will be defining a new system of geometry using complex vectors and introducing dimensions as sets. This will be done through a geometric concept of sets of dimensions that sum to an n -dimensional space from $n \rightarrow -\infty$ to $n \rightarrow \infty$. This paper is **VERY** rough and a collection of thoughts I am working on, when it is done it may be a small book with properly composed chapters.

I would also like to dedicate this paper to those who have inspired me: my God, Jesus, my mother, and my teacher.

Notes on term usages

The following groups of terms will be used interchangeably, where the proper term is in bold:

1. Space, Geometric Space, **N-Dimensional (Geometric) Space**.
2. Axis, **Rigid Body**.
3. Transformations, Transformation Functions, **Geometric Transformation Functions**.

Definition of ε (The Smallest Number) and χ_∞

We will now define ε as the limit of $1/x$ as $x \rightarrow \infty$. When this is seen in a function of equation, which we will call $E(x)$, it is the same as taking the limit of $E(x)$ as $\varepsilon \rightarrow 0$; $\lim_{\varepsilon \rightarrow 0} E(\varepsilon)$.

This is the “Smallest Number” such that $2\varepsilon = \varepsilon$, while maintaining a linear rate of “increase”. This number can be thought of as the difference between $0 < \varepsilon$ as $\varepsilon \rightarrow 0$.

Now we establish the largest number, we will call it χ_∞ , as the quasi-transfinite number. It is the cardinality of transfinite numbers. Formally: $\chi_\infty = \lim_{x \rightarrow \inf} \aleph_x$.

Definition of a Point and The Universal Origin

To begin we must first define what a point is. A spatial vector is a vector in the form of $V = \{d_k | d \in \mathbb{C}\}_{k \in \mathbb{R}}$ where d is variable and k is fixed index for all vectors. A point is the fundamental unit of all geometric spaces. A point has an n -dimensional n value of the sum of all of its vector fields $n = \sum_{d \in V} d$.

We will now define the **Universal Origin**, represented by the archaic Greek letter koppa (Q_U), as $Q_U = \{d_k | d = 0\}_{k \in \mathbb{Z}}$. We denote the origin of a geometric shape A , with Q_A , and the every point as Q_V where V is a co-ordinate vector. A geometric space's origin is denoted as $Q_{Q[D]}$. A co-ordinate vector is a spatial vector used for co-ordinates with respect to an origin in a space.

Dimensions as Spatial Densities

The **Dimensional Density Set** of a space is denoted as D . A space is defined by $S = Q[D]$ where D is the vector of the dimensions of S . A space's dimensions are the complex densities of a **Inclusive Spatial Area** (Defined later). In traditional geometries these would be 1 for n many fields of D . The n -dimensionality of a space is defined as the sum of all elements of D ; $\sum_{d \in D} d$. This sum is can be generalized in some cases, such as in 2d euclidean geometry, $D = \{1, 1\}$ implies that $n=2$. This is in-itself not the same though, its a simple framework for interpretation. It does not capture the full complexity of dimensionality.

We must now ask what dimensionality means. In order to understand this, we must stop thinking about dimensions as lines but as purely complex numbers, i.e. we will think of them as vectors. In this sense dimensionality is a spectrum of density, such that along any given "axis" space may be more dense than along another. This is due to the fact that the value of a vector's field is only the root d of the intrinsic area where d is the dimensional density, i.e. $a=f^d$.

The way that this is derived is by the grouping of powers of factored dimensions from areas of euclidean spaces, and further extending them into an two infinite sets of powers and bases. This is where we derive the meaning of dimensionality. This then allows for the extension into complex numbers and negative spaces.

The application of these principles will be shown later on, but are built off of these principles. If it was unclear, it is advised to re-read these paragraphs and/or the definition of D at the end of this paper.

Space and $\mathfrak{N}$

A space is defined by $S = Q[D]$ where D is the vector of the dimensions of S . A space's dimensions are the complex densities of a **Inclusive Spatial Area** (Defined later). In traditional geometries these would be 1 for n many fields of D . The n -dimensionality of a space is defined as the sum of all elements of D ; $\sum_{d \in D} d$.

A space is merely a set of three sets. The set that distinguishes this space from other spaces, the set of dimensional densities in this space. Then there are the two universally shared sets: the set of all possible vector sets (P) and the set of all possible shapes, i.e. the set of all subsets of P .

We will also define a geometric space that is the combined existence of all spaces. This space we will call $\mathfrak{N}$. In this space every the D fields are sets of all valid values for each field, thus it is an space where all operations are equally valid in parallel. This space has a inclusive spatial area of χ_{∞} .

Definition of A Line

A line is defined by Q_{V1} and Q_{V2} where $V1 \neq V2$, where $V2$ and $V2$ are vectors. A point's vector has infinity many fields.

Geometric Transformation Functions

Here I will briefly touch on the systems required for the transformation of shapes and vectors between unlike spaces. This is an issue due to the fact that a space's vectors may not have a direct correspondence, e.g. in sine curved spaces or spaces of varying values of D fields.

These systems are a sets of functions to transfer vectors between geometries. They are always relative from geometry A to geometry B . There are infinitely many forms of these spaces but here I will touch on a sine space.

First we will label its dimensional set fields x_s and y_s , and the Euclidean equivalent geometric dimensional set fields x_E and y_E . Now we define the functions: $y_E = \sin(y_s)$ and $x_E = x_s$. The generalized definition of Geometric Transformation Functions is $k_{s2} = f_k(k_{s1})$, where k is an index to D and V . This is a simple form for 2d spaces with vector shifting. There are also more complex forms that shift between multiple dimensions and do not use the $k_{s2} = f_k(k_{s1})$ formula. $k_{s2} = f_k(k_{s1})$ only applies as a generalized form of 2d shifts in vector field arrangement, such as dimensional densities.

One notable property is that a transformation function may shift a vector field into a field that is "hidden" in one geometry, thus causing a lowering of I_o , while not changing the shape itself. This is most notable in a transformation of a sphere between a $D = \{\dots[0], 1, 1, 1, [0]\dots\}$ and a $D = \{\dots[0], 1, 1, [0]\dots\}$ space. Here we hide one dimensional field resulting in a euclidean 2d circle state of a 3d sphere.

Shapes

Here we will talk about shapes. A shape shall be denoted with the symbol O possibly with a subscript representing its origin relative to Q_s and its upper vector.

A shape has an **Inclusive Spatial Area**. This area is a number which represents the the includes area from of a shape's origin to it's upper vector. This value is represented with an I it is calculated with a function denoted as $I(O, D)$.

As the upper vector will always be absolute value of a complex number with positive coefficients in the form $a+bi$, we know that the area will always be a real number. A shape's origin shall be denoted as Q_O where O is the shape. This must be absolute as if it isn't then in many spaces there will be alternation between negative/imaginary values of I and positive, depending on the density ratio and relative vector fields.

Equations and Elements

Here I will explain the equations and the elements used in this geometry. This is meant as a in depth reference.

$$k \in \mathbb{R}$$

k represents any index in a vector set. This is designed to match the cardinality of $\mathbb{R}$ i.e. $2^{\aleph_0}$, so that it may index D and V vector sets.

$$D = \{x_k | x_k \in \mathbb{C}\}$$

Here we define the structure of a space's dimension density set (D), to have density elements indexed by k . This is simply a prototype structure and not a definite set, that varies per space.

$$r_k \in D$$

Here we define r_k as an element of a set using the prototype set D .

$$|D| = 2^{\aleph_0} = \beth_1$$

Here we establish the cardinality of any set using the prototype set D . Note how this is the same as the cardinality the set k originates from.

$$V = \{x_k | x_k \in \mathbb{C}\}$$

Here we define the structure of a vector set, where dimensional values are indexed by k .

This is a prototype structure like D and varies per vector set.

$$d_k \in V$$

Here we define d_k as an element of a set using the prototype set D .

$$|V| = |D|$$

Here we establish the cardinality of a vector set as equivalent to the cardinality of D .

$$P(V, D) = \{(d_k, r_k)\}$$

Here we define the function $P(V, D)$ that when given a vector set and dimension density set, resolves into a set of pairs of the two sets joined by their index k .

$$r_k \Rightarrow d_k$$

Here we establish that d_k implies the existence of r_k .

$$n = \sum_{r \in D} r$$

Here we show the equation for the n value of n -dimensional space. Note that this does not fully capture the complexity of a space.

$$S = Q[D]$$

Here we show the notation for a space's set S , i.e. we use the kappa space notation to define a space's set of all geometric objects (vector sets, and shapes), as a set of all geometric objects that exist in space.

$$O \in S$$

Here we define O as an object in space S .

$$I(O_{V_d}, D) = \prod_{(d_k, r_k) \in P(O_v, D)} \lim_{x=d_k} x^{r_k}$$

Here we define the function that when given a shape's dimensions (as a vector set of the outer vector with respect to a shape's origin (Q_0)) and a space's dimensional density set, resolves to the **Inclusive Vector Area**, with respect to the space it resides in.